

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and Communications Technology

Software Requirement Specification

Version 1.2

AIMS Project

Subject: ITSS Software Development

Group 15

Lương Ngọc Vũ Long – 20235967

Phạm Thanh Phong – 20235987

Cao Văn Đông - 20235911

Nguyễn Ngọc Minh Hiếu - 20235933

Trịnh Lương Việt - 20236012

Hanoi, 3/2026

Table of contents

Table of contents	1
1 Introduction.....	2
1.1 Objective.....	2
1.2 Scope	2
1.3 Glossary	2
1.4 References.....	3
2 Overall Description.....	4
2.1 Survey.....	4
2.2 Overall requirements	4
2.3 Business process	4
3 Detailed Requirements.....	5
3.1 Use case 1	6
3.2 Use case 2	7
4 Supplementary specification.....	8
4.1 Functionality	8
4.2 Usability.....	8
4.3 Reliability	8
4.4 Performance.....	8
4.5 Supportability	8
4.6 Other requirements	8

1 Introduction

1.1 Objective

The primary purpose of this Software Requirement Specification (SRS) is to establish and maintain a formal agreement between the customers and the developers on what the AIMS (An Internet Media Store) system should do. This document provides a complete and unambiguous description of the functional and non-functional requirements for the AIMS software, which is defined as an e-commerce software (website or desktop app).

The SRS specifically outlines the requirements for the buying and selling of physical media products, including books, newspapers, CDs, and DVDs. It serves as a technical foundation for the following intended audiences:

- Analysts and Designers: To provide a structural basis for the Software Architecture Design and Software Detailed Design processes.
- Developers: To ensure a clear understanding of the system's logic.
- Testers: To establish qualification requirements and criteria to verify that the software.
- Project Managers: To provide a basis for estimating cost, time and planning the technical contents of the development iterations.

1.2 Scope

The software product to be produced is AIMS (An Internet Media Store), an e-commerce software available as either a website or a desktop application.

What the software will do: The system is designed to facilitate the buying and selling of physical media products, specifically books, newspapers, CDs, and DVDs. Key functionalities include:

- For Customers: Browsing a 20-product random list, searching by title or category, filtering by price, managing a shopping cart, and placing orders with calculated shipping fees.
- For Product Managers: Adding, editing, and deleting products, manually adjusting stock with recorded reasons, and reviewing or rejecting pending orders.
- For Administrators: Managing users (creating, updating, deleting, blocking), resetting passwords, and assigning multiple roles to users.
- Payment & Security: Integration of VietQR and PayPal Sandbox for transactions, while maintaining business rules such as the 30%–150% price constraint to avoid inflation.

What the software will not do: Currently, AIMS supports only physical media products; other types of products or digital media services are not supported in the current version. Additionally, while it supports QR codes and credit cards, other payment methods and notification channels (like SMS) are reserved for future updates.

Objectives and Benefits: The primary goal is to provide a platform that operates 24/7 and allows new users to easily familiarize themselves. The software is engineered for high performance and reliability, capable of serving 1,000 customers simultaneously with a maximum response time of 2 normally and up to 5 seconds during peak hours. It aims for high availability, operating for 300 hours without failure and ensuring recovery within 1 hour after any incident

1.3 Glossary

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
1	AIMS	A software system called <i>An Internet Media Store</i> used to buy and sell physical media products such as books, CDs, DVDs, and newspapers.	AIMS website or desktop app	Main system described in this document
2	Barcode	A unique number used to identify each product in the system.	893523523	Must be different for every product
3	Original Price	The base price of a product before any adjustments or taxes are applied.	100,000 VND	Used for price comparison
4	Current Price	The actual selling price of a product in the system.	120,000 VND	Must follow system rules of between 30%-150% of original price
5	Payment Gateway	An external system that processes payments securely.	PayPal	Handles transactions including payment and refund
6	VAT (Value-Added Tax)	A 10% tax added to product prices during payment.	10% VAT	Not included in stored prices

1.4 References

The following documents and resources have been used as references in the preparation of this Software Requirements Specification:

[1] M. Fowler and K. Scott, *The Unified Modeling Language Reference Manual*, 2nd edition, Addison-Wesley, 2004.

[2] Hanoi University of Science and Technology, *ITSS Software Development Course Materials*, 2026.

[3] *AIMS Project Specification*, ITSS Software Development Course, Hanoi University of Science and Technology, 2026.

[4] PayPal, *PayPal Developer Documentation*. Available: <https://developer.paypal.com/>

[5] VietQR, *VietQR Payment Documentation*. Available: <https://api.vietqr.vn/>

2 Overall Description

2.1 Survey

AIMS is an e-commerce software (available as a website or desktop application) that operates 24/7 to facilitate the trade of physical media. The system is designed to be user-friendly for new users while supporting high-concurrency for up to 1,000 simultaneous customers.

The system interacts with the following three primary actors:

- Customer: An unauthenticated user who interacts with the system to browse a random list of products, search for items by title or category, and filter results by price. Customers manage their own shopping carts, provide delivery information, and perform payments via QR codes or credit cards. They can also cancel orders before they are approved by a manager.
- Product Manager: An internal staff member responsible for the lifecycle of products and orders. They have the authority to add, edit, or delete products and must manually adjust stock levels for reasons like damage or new arrivals. Additionally, they are responsible for reviewing, approving, or rejecting pending orders.
- Administrator: A high-level system user responsible for user management. Their duties include creating new user accounts, updating user information, resetting passwords, and blocking or unblocking users as necessary. They also define system access by setting or changing user roles, noting that a single user can hold multiple roles simultaneously.

2.2 Overall requirements

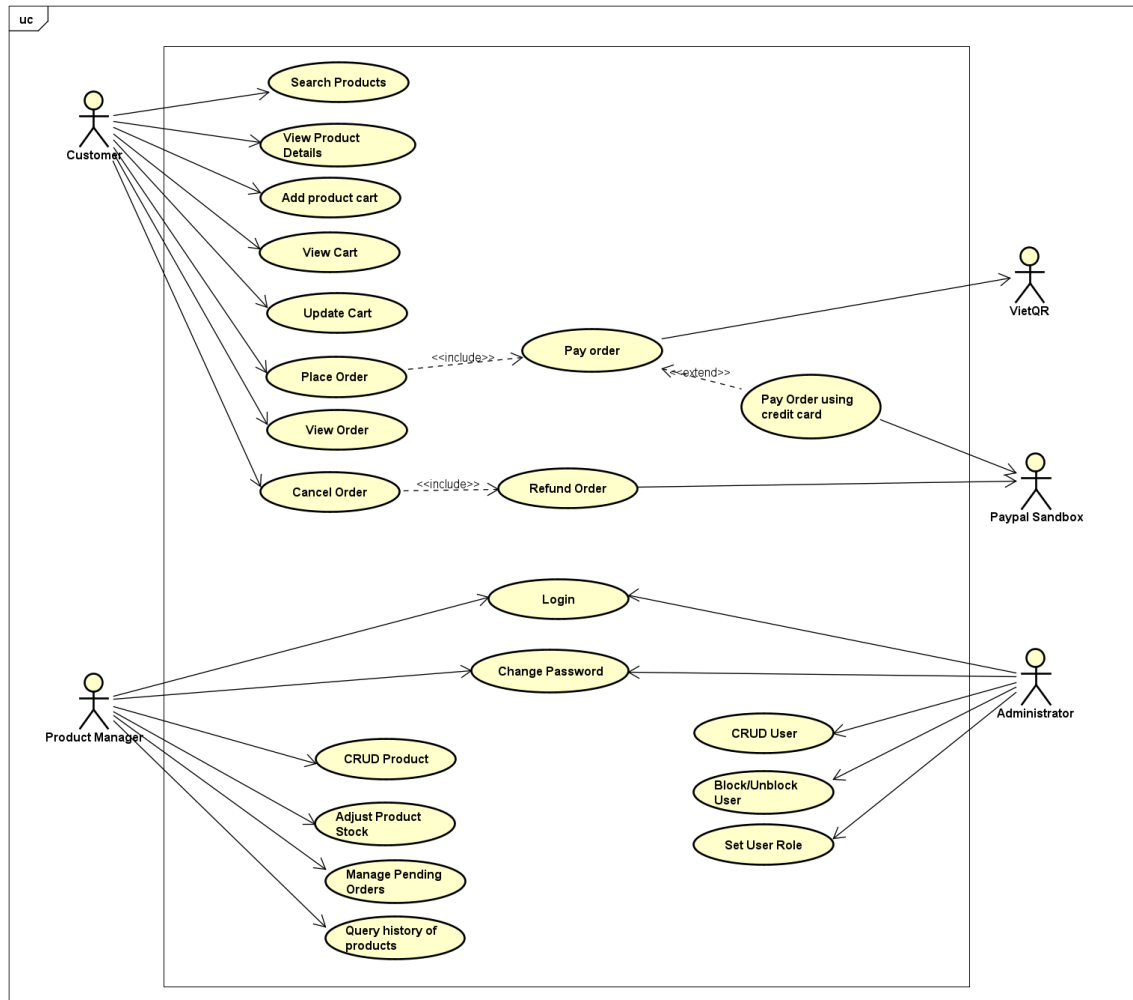


Figure 1: Overall Use Case Diagram

2.3 Business process

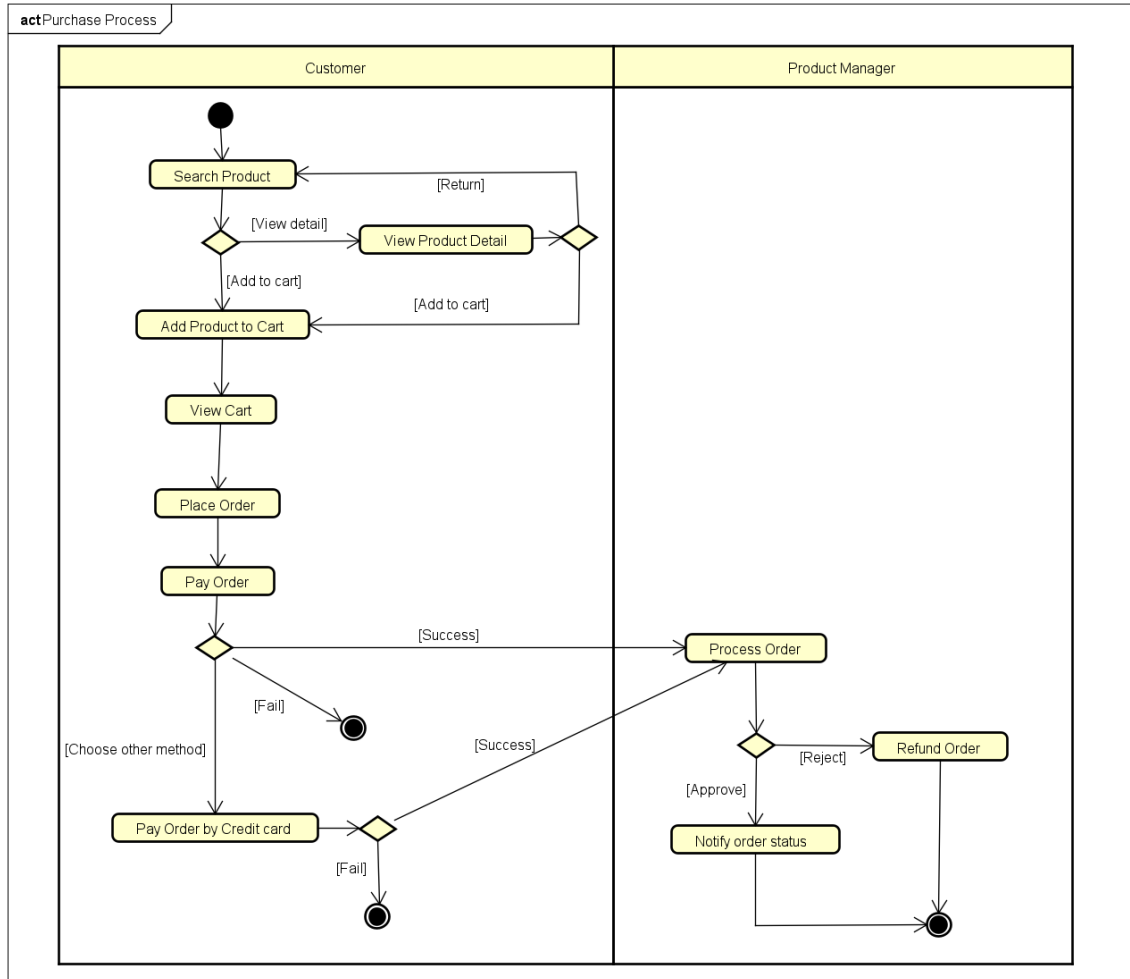


Figure 2: Purchase Process

3 Detailed Requirements

3.1 Use case “Add Product”

3.1.1. Use case code

UC001

3.1.2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS Software when the Product Manager wish(es) to add a new media product (Book, Newspaper, CD, or DVD) to the store. The Product Manager must provide essential information, physical attributes, and core details for each item.

3.1.3. Actors

- Product Manager

3.1.4. Preconditions

- The Product Manager has successfully logged into the AIMS Software with their designated account and role
- The system is at the product management interface and is ready to receive new product information

3.1.5. Basic Flow of Events

1. The Product Manager initiates the request to add a new media product.
2. The AIMS Software prompts for the product's core details, physical attributes, and essential information.
3. The Product Manager enters the common product information (see Table 1) and selects the specific media category.
4. The AIMS Software prompts for further details based on the selected category (Book, CD, DVD, or Newspaper).
5. The Product Manager enters the specific details and any additional information corresponding to the category (see Table 2, 3, 4, 5, 6).
6. The AIMS Software validates the business rules, ensuring the current price is between 30% and 150% of the original value and the date format is correct.
7. The AIMS Software saves the new product record to the database.
8. The AIMS Software records the transaction details (including Manager ID and timestamp) in the history of products addition.
9. The AIMS Software displays a success notification.

3.1.6. Alternative flows

Table 1: N-Alternative flows of events for UC Add Product

No	Location	Condition	Action	Resume location
1.	At Step 6	If the current price is not between 30% and 150% of the	• The AIMS Software notifies the product manager that the operation is invalid (see Table B).	Resumes at Step 3

		product's original value.		
2.	At Step 6	If the date format (e.g., for publication or release dates) is incorrect.	The AIMS Software notifies the product manager that the operation is invalid (see Table B).	Resumes at Step 3
3.	At Step 6	If the input be blank.	The AIMS Software notifies the product manager that the operation is invalid.	Resumes at Step 3
4.	Anytime	The manager cancels the edit.	The system does not save changes and closes the form.	Use case ends

3.1.7. Input data

Table 2-Input data of Common media details

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Barcode	Part of the core details for product identification.	Yes	Unique identifier; must not be empty.	893523523
2.	Title	Part of the core details.	Yes	Must not be empty.	MIDNIGHT FOREVER EXPO 'MEIKŌ WA GUNARUGA GOTOSHI'
3.	Category	Part of the core details (e.g., Book, Newspaper, CD, DVD).	Yes	Must be Book, Newspaper, CD or DVD	DVD
4.	Original value	Part of the core details. Price is without a 10% value-added tax (VAT).	Yes	Numerical and positive value.	1,765,420 VNĐ
5.	Current price	Part of the core details. Price is without a 10% value-added tax (VAT).	Yes	Must be between 30% and 150% of the product's original value.	1,823,234 VNĐ
5	Width	Width of product	Yes	Numerical and positive.	5
6	Length	Length of product	Yes	Numerical and positive.	6

7	Height	Height of product. (cm)	Yes	Numerical and positive value.	7
8	Weight	Part of the physical attributes (kg)	Yes	Numerical and positive value.	0.2
9	General description	Essential information such as new or used condition, primary color, or return condition.	Yes	Must provide a clear description of item status.	New condition, original Blu-ray package, Japanese concert recording, non-returnable

Table 3 -Input data of Specific details for Books

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Author(s)	Further details required for books.	Yes	Must not be empty.	ZUTOMAYO
2.	Cover type	Further details required for books.	Yes	Must be paperback or hardcover.	hardcover
3.	Publisher	Further details required for books.	Yes	Must not be empty.	Universal Music LLC
4.	Publication date	Further details required for books.	Yes	Must follow the correct date format.	29/11/2023
5.	Number of pages	Additional information that may be included.	No	Numerical and positive integer.	160
6.	Language	Additional information that may be included.	No	String text.	Japanese
7.	Genre	Additional information that may be included.	No	String text.	Photobook

Table 4-Input data of Specific details for CDs

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Artists	The performers or creators of the music.	Yes	Not empty.	ZUTOMAYO

2.	Record labels	The company that produced the CD.	Yes	Not empty.	EMI Records / Universal Music LLC
3.	Tracks list	List of songs, each with its own title and length.	Yes	Must include title and duration.	Aizura Zen'in Dousoukai - 4:14
4.	Genre	The category of music.	Yes	Not empty.	J-Pop
5.	Release date	The date the CD was released.	No	Correct date format.	18/06/2021

Table 5-Input data of Specific details for DVDs

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Disc type	The format of the DVD.	Yes	Blu-ray or HD-DVD only	Blu-ray
2.	Director	The person who directed the film.	Yes	Not empty.	ZUTOMAYO
3.	Runtime	Total length of the video.	Yes	Numerical and positive value.	121 (mins)
4.	Studio	The production company.	Yes	Not empty.	Universal Music LLC
5.	Language	The primary audio language.	Yes	Not empty.	Japanese
6.	Subtitles	Available subtitle languages.	Yes	Not empty.	Japanese, English, Chinese
7.	Release date	The date the DVD was released.	No	Correct date format.	25/09/2024
8.	Genre	The film category.	No	String text.	Concert / J-Pop

Table 6-Input data of Specific details for Newspapers

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Editor-in-chief	The person in charge of the newspaper's content.	Yes	Must not be empty.	Hiroyuki Nakano
2.	Publisher	The company that publishes the newspaper.	Yes	Must not be empty.	Shueisha
3.	Publication date	The date the newspaper was published.	Yes	Must follow the correct date format.	15/03/2026
4.	Issue number	The specific identification number of the issue.	No	Numerical or string format.	No. 15 (2026)
5.	Publication frequency	How often the newspaper is published (e.g., daily, weekly).	No	String text.	Weekly
6.	ISSN	International Standard Serial Number for serial publications.	No	Standard ISSN format.	2049-3630
7.	Language	The primary language used in the newspaper.	No	String text.	Japanese
8.	Sections	Specific parts of the newspaper (e.g., politics, business, sports, culture).	No	String text; list of categories.	Manga, Action, Comedy, Sports, Fantasy

3.1.8. Output data

Table 7 -Output data of Add Product

No	Data fields	Description	Display format	Example
1.	Success notification	Message confirming that the product has been successfully added to the system.	Text/Pop-up	Product added successfully.
2.	Operation invalid notification	Message notifying the manager if the input violates rules (30%-150% price rule) or if the date format is incorrect.	Text (Red) or Pop-up	Invalid current price: must be between 30% and 150% of the original value.
3.	History of products addition	A confirmation record stored in the system history	Status message/System Log	[Date] [Time] - Product [Barcode] added by [Manager ID].

		reflecting the addition.		
4.	Product summary	A display of the core details of the newly added product for verification.	Text/Table (Current price-aligned with VND)	Title: MIDNIGHT FOREVER EXPO ‘MEIKŌ WA GUNARUGA GOTOSHI’ ; Price: 1,828,234VND.

3.1.9. Postconditions

- If the operation is successful: The new product information is stored in the database, and the transaction is recorded in the history of products addition.
- If the operation is cancelled or invalid: The system state remains unchanged; no product data is updated, and no entry is added to the history log.

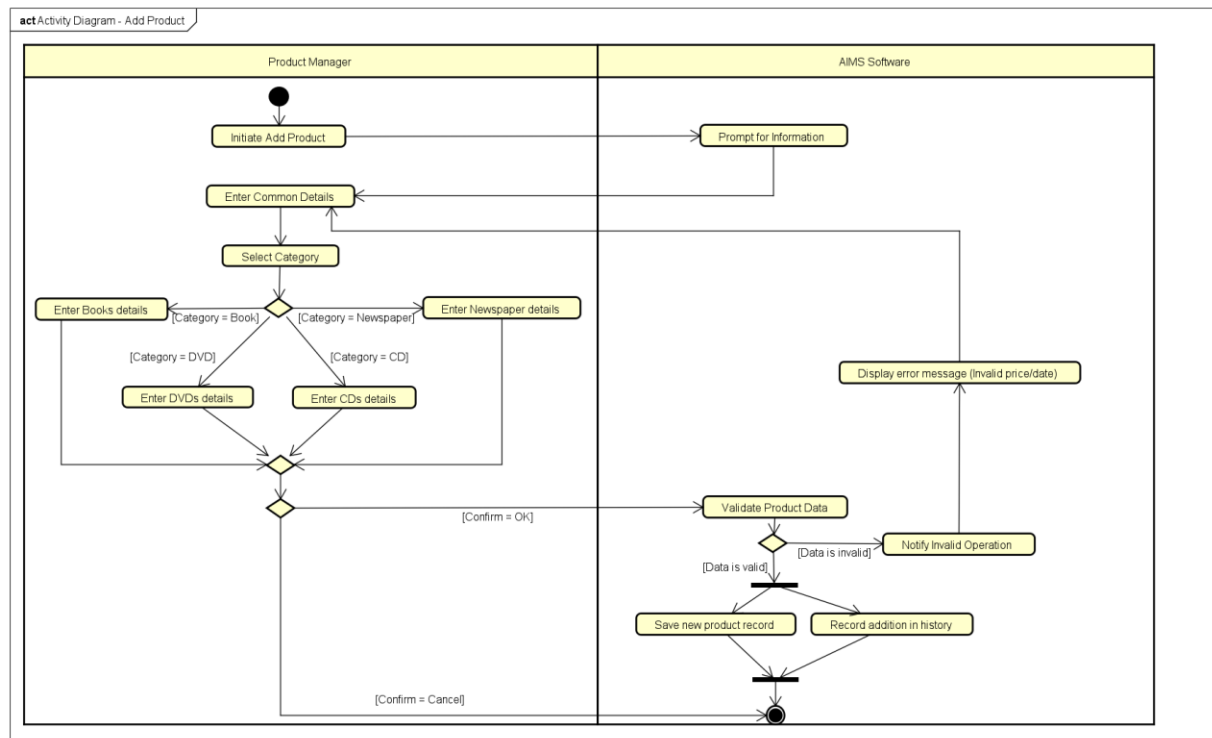


Figure 3: Activity Diagram – Add Product

3.2 Use case “Edit Product”

3.2.1. Use case code

UC002

3.2.2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS Software when the Product Manager wish(es) to edit the information of an existing media product (Book, Newspaper, CD, or DVD). The Product Manager can update core details, physical attributes, essential information, or specific category details, but can only edit one product at a time

3.2.3. Actors

- Product Manager

3.2.4. Preconditions

- The Product Manager has successfully logged into the AIMS Software with their designated account and role.
- The system is at the product management interface and a list of existing products is displayed.
- The specific product intended for editing has been selected by the Product Manager

3.2.5. Basic Flow of Events

1. The Product Manager selects one and only one existing media product from the list and requests to edit its information.
2. The AIMS Software retrieves and displays all current information of the selected product, including core details, physical attributes, and category-specific details.
3. The Product Manager modifies the data fields. Note: Both original value and current price must be entered without the 10% value-added tax (VAT).
4. The AIMS Software validates the updated information:
 - a. Price Rule: Ensures the current price is between 30% and 150% of the product's original value.
 - b. Format Rule: Verifies the date format and ensures all mandatory fields are correctly provided.
5. The AIMS Software updates the product record in the database.
6. The AIMS Software records the transaction details (including Manager ID, timestamp, and specific changes) in the history of products editing.
7. The AIMS Software displays a success notification to the Product Manager

3.2.6. Alternative flows

Table 8: N-Alternative flows of events for UC Edit Product

No	Location	Condition	Action	Resume location
1.	At Step 4	If the current price is outside the 30%-150% range of the original value.	▪ The software notifies the manager that the operation is invalid.	Resumes at Step 2
2.	At Step 4	If the date format is incorrect or mandatory fields are missing.	▪ The software notifies the manager that the operation is invalid.	Resumes at Step 2
3.	Anytime	The manager cancels the edit.	▪ The system does not save changes and closes the form.	Use case ends

3.2.7. Input data

Table 9: Input data of Common media details

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Barcode	Part of the core details for product identification.	Yes	Unique identifier; must not be empty.	893523523
2.	Title	Part of the core details.	No	Must not be empty.	MIDNIGHT FOREVER EXPO 'MEIKŌ WA GUNARUGA GOTOSHI'
3.	Category	Part of the core details (e.g., Book, Newspaper, CD, DVD).	No	Must be a supported media type.	DVD
4.	Original value	Part of the core details. Price is without a 10% value-added tax (VAT).	No	Numerical and positive value.	1,765,420 VNĐ
5.	Current price	Part of the core details. Price is without a 10% value-added tax (VAT).	No	Must be between 30% and 150% of the product's original value.	1,823,234 VNĐ
6.	Width	Width of product	Yes	Numerical and positive	5
7.	Length	Length of product	Yes	Numerical and positive.	6
8.	Height	Height of product. (cm)	Yes	Numerical and positive value.	7
9.	Weight	Part of the physical attributes.	No	Numerical and positive value.	0.2 (kg)
10.	General description	Essential information such as new or used condition, primary color, or return condition.	No	Must provide a clear description of item status.	New condition, original Blu-ray package, Japanese concert recording, non-returnable

Table 10 -Input data of Specific details for Books

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Author(s)	Further details required for books.	No	Must not be empty.	ZUTOMAYO
2.	Cover type	Further details required for books.	No	Must be paperback or hardcover.	hardcover
3.	Publisher	Further details required for books.	No	Must not be empty.	Universal Music LLC
4.	Publication date	Further details required for books.	No	Must follow the correct date format.	29/11/2023
5.	Number of pages	Additional information that may be included.	No	Numerical and positive integer.	160
6.	Language	Additional information that may be included.	No	String text.	Japanese
7.	Genre	Additional information that may be included.	No	String text.	Photobook

Table 11 -Input data of Specific details for CDs

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Artists	The performers or creators of the music.	No	Not empty.	ZUTOMAYO
2.	Record labels	The company that produced the CD.	No	Not empty.	EMI Records / Universal Music LLC
3.	Tracks list	List of songs, each with its own title and length.	No	Must include title and duration.	Aizura Zen'in Dousoukai - 4:14
4.	Genre	The category of music.	No	Not empty.	J-Pop
5.	Release date	The date the CD was released.	No	Correct date format.	18/06/2021

Table 12 -Input data of Specific details for DVDs

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Disc type	The format of the DVD.	No	Must be Blu-ray or HD-DVD.	Blu-ray
2.	Director	The person who directed the film.	No	Not empty.	ZUTOMAYO
3.	Runtime	Total length of the video.	No	Numerical and positive value.	121 (mins)
4.	Studio	The production company.	No	Not empty.	Universal Music LLC
5.	Language	The primary audio language.	No	Not empty.	Japanese
6.	Subtitles	Available subtitle languages.	No	Not empty.	Japanese, English, Chinese
7.	Release date	The date the DVD was released.	No	Correct date format.	25/09/2024
8.	Genre	The film category.	No	String text.	Concert / J-Pop

Table 13-Input data of Specific details for Newspapers

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Editor-in-chief	The person in charge of the newspaper's content.	No	Must not be empty.	Hiroyuki Nakano
2.	Publisher	The company that publishes the newspaper.	No	Must not be empty.	Shueisha
3.	Publication date	The date the newspaper was published.	No	Must follow the correct date format.	15/03/2026
4.	Issue number	The specific identification number of the issue.	No	Numerical or string format.	No. 15 (2026)
5.	Publication frequency	How often the newspaper is published (e.g., daily, weekly).	No	String text.	Weekly
6.	ISSN	International Standard Serial Number for serial publications.	No	Standard ISSN format.	2049-3630

7.	Language	The primary language used in the newspaper.	No	String text.	Japanese
8.	Sections	Specific parts of the newspaper (e.g., politics, business, sports, culture).	No	String text; list of categories.	Manga, Action, Comedy, Sports, Fantasy

3.2.8. Output data

Table 14-Output data of Edit product

No	Data fields	Description	Display format	Example
1.	Pre-filled Product Form	The software retrieves and displays the current data of the selected Album from the database onto the edit form.	Form fields	Title: MIDNIGHT FOREVER EXPO ‘MEIKŌ WA GUNARUGA GOTOSHI’; Artist: ZUTOMAYO; Category: CD; Current Price: 1,828,234 VND
2.	Success notification	Confirmation message shown after the manager clicks "Save" and all data complies with business rules.	Text / Pop-up	Product updated successfully!
3.	Operation invalid notification	Error message displayed if the new price is outside the 30% - 150% range of the original value or if input is invalid.	Text (Red)	Invalid price: New price (500,000) is below 30% of original value.
4.	History of products editing	The system automatically records the transaction details, including the manager ID and changes made, into the audit log.	System Log	[2026-03-16] Manager [ID:01] updated [Current Price] of Album [Barcode: 893523523] from 1,828,234 to 2,000,000 VND
5.	Updated Product Details	A summary screen or table showing the final updated information of the Album for verification.	Table / Screen	Album: Hisohiso Banashi; Updated Price: 2,000,000 VND; Status: Updated

3.2.9. Postconditions

- If the operation is successful: The product information is updated in the database, and the transaction is recorded in the history of products editing.
- If the operation is cancelled or invalid: The system state remains unchanged; no product data is updated, and no entry is added to the history log.

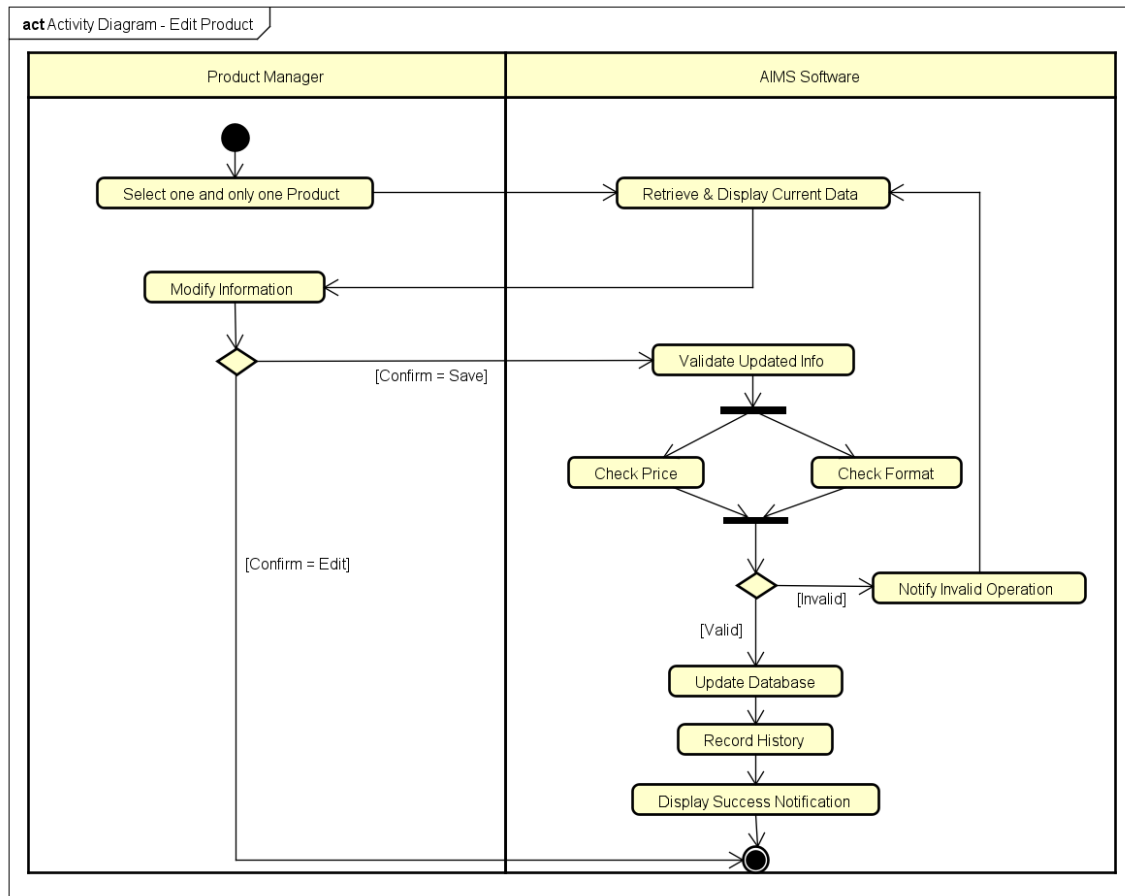


Figure 4: Activity Diagram – Edit Product

3.3 Use case “Delete Product”

3.3.1. Use case code

UC003

3.3.2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS Software when the Product Manager wish(es) to remove existing media products from the system. The software allows deleting up to ten products simultaneously. Depending on the stock level, the system will either permanently delete the product or change its status to "deactivated"

3.3.3. Actors

- Product Manager

3.3.4. Preconditions

- The Product Manager has successfully logged into the AIMS Software with their designated account and role.
- The system is at the product management interface and a list of existing products is displayed.
- The Product Manager has selected at least one (and up to ten) products for deletion

3.3.5. Basic Flow of Events

1. The Product Manager selects at least one and up to ten existing media products from the product list and requests to delete them.
2. The AIMS Software verifies the daily deletion limit, ensuring the manager has not exceeded the maximum of twenty products per day.
3. For each selected product, the AIMS Software checks the current stock quantity.
4. The AIMS Software processes the removal based on the following stock conditions:
 - a. If stock is equal to zero: The software permanently removes the product record from the active sales list.
 - b. If stock is greater than zero: The software does not delete the product but changes its status to "deactivated", making it unavailable for sale.
5. The AIMS Software records the transaction details in the history of products deletion.
6. The AIMS Software displays a notification to the Product Manager, summarizing the number of products successfully deleted and those moved to "deactivated" status.

3.3.6. Alternative flows

Table 15: N-Alternative flows of events for UC Delete Product

No	Location	Condition	Action	Resume location
1.	At Step 1	The Product Manager selects more than ten products.	The software notifies the manager that the operation is invalid because the selection exceeds the 10-product limit per request,.	Resumes at Step 1
2.	At Step 2	The deletion request would cause the manager to exceed the limit of twenty products per day.	The software notifies the manager that the operation is invalid for security reasons and the daily limit has been reached,.	Use case ends
3.	Anytime	The manager cancels the deletion.	The system does not perform any deletion or status changes. The system state remains unchanged.	Use case ends

3.3.7. Input data

Table 16-Input data of Common media details

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Selected Product List	List of unique identifiers (barcodes) of the media products the	Yes	Selecting between 1 and 10 products per operation.
- Total deleted	

		manager wants to remove.		products must not exceed 20 per day for the manager.	
--	--	--------------------------	--	--	--

3.3.8. Output data

Table 17-Output data of Edit product

No	Data fields	Description	Display format	Example
1.	Deletion Summary Notification	Display a summary notification after processing the product list (the actual number of products deleted and the number of products that were deactivated).	Text / Pop-up	Successfully deleted 3 products and deactivated 2 products (due to remaining stock).
2.	Invalid Operation Notification	Display an error notification when the Manager selects more than 10 products or exceeds the limit of deleting 20 products per day.	Text (Red)	Operation invalid: Daily deletion limit (20) has been reached.
3.	History of products deletion	The system automatically records detailed logs of delete/deactivation transactions for auditing purposes.	System Log	[2026-03-16] Manager [ID:01] deleted 3 products and deactivated 2 products.
4.	Updated Product List	The product list is refreshed on the management screen after the operation is completed.	Table / Screen	

3.3.9. Postconditions

- If the operation is successful: The product is permanently removed (if stock is zero) or its status is changed to “deactivated” (if stock is greater than zero) in the database, and the transaction is recorded in the history of products deletion
- If the operation is cancelled or invalid: The system state remains unchanged; no product data is updated, and no entry is added to the history log

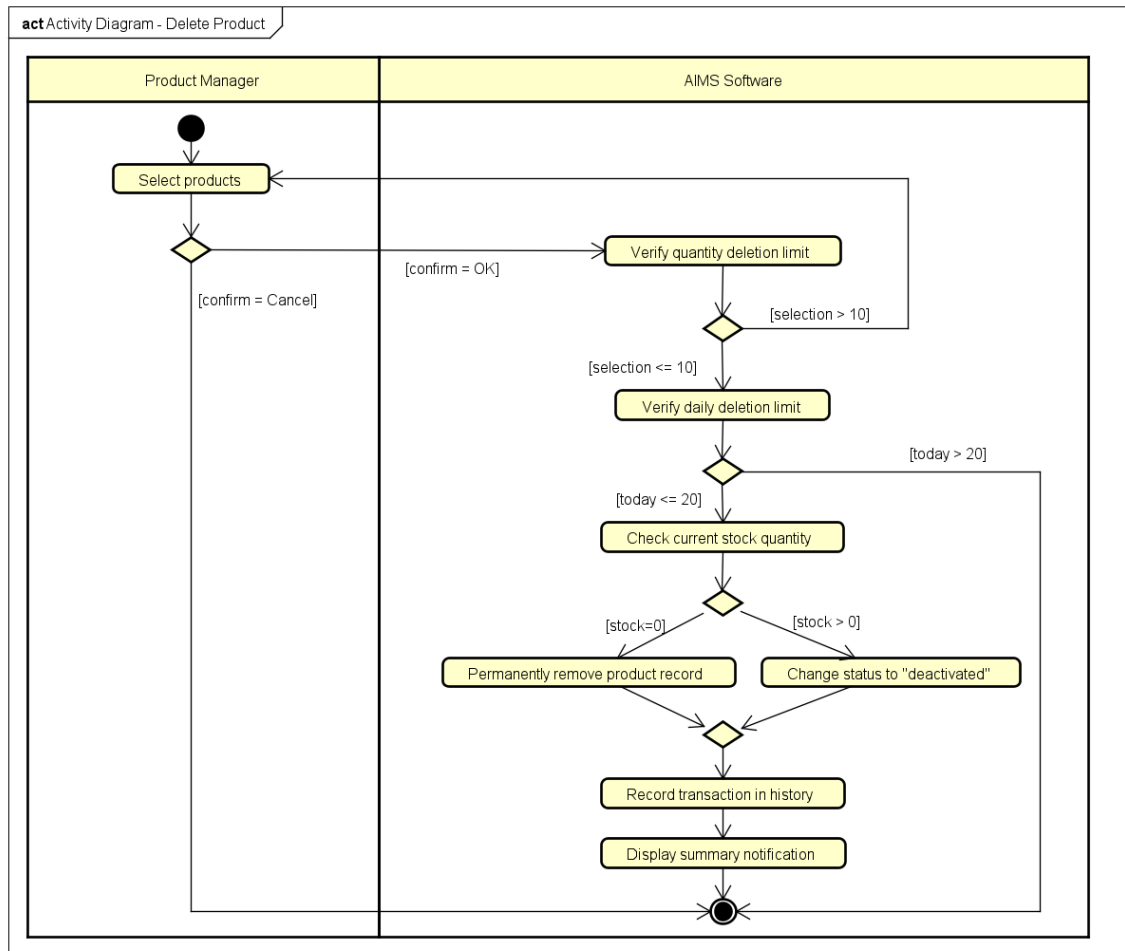


Figure 5: Activity Diagram – Delete Product

3.4 Use case “Place Order”

3.4.1 Use case code

UC004

3.4.2 Brief Description

This use case describes the interaction between the Customer and the AIMS system when the Customer wishes to convert items in their shopping cart into a successfully paid order with delivery information.

3.4.3 Actors

- Customer

3.4.4 Preconditions

- The Customer has at least one item in the shopping cart.
- The Customer is at the "View Cart" screen

3.4.5 Basic Flow of Events

1. Customer requests to place order from the cart view.
2. Software checks whether the available stock is sufficient for all items in the cart.
3. Software displays delivery information form (see Table 12).

4. Customer enters delivery information.
5. Software checks for delivery information validation.
6. Software calculates and displays the shipping fee.
7. Software displays and temporarily saves invoice information (see Table 13).
8. Customer requests to pay the order.
9. Software calls the use case **"Pay Order"**.
10. Software displays general order and transaction information(see Table 13).
11. Software updates the order state to "Pending".
12. Software sends a confirmation email.
13. Software records the payment transaction and the successfully paid order.
14. Software displays the payment transaction and the successfully paid order.

3.4.6 Alternative flows

Table 18-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	If software detects insufficient stock for items	Software informs customer of the lacking product.	Use case end
2.	At Step 5	If software detects invalid delivery information	Software prompts user for corrections.	Resumes at Step 4
3.	At Step 7	Customer requests to adjust delivery address	Software allows customer to return to Step 4	Resumes at Step 4
4.	At Step 9	Payment is canceled or fails in "Pay Order" use case	Software informs the customer of the payment status before returns to the invoice summary.	Resumes at Step 7
5.	Any Step	If customer requests to exit or go back	Software cancels current process and returns to previous view or main view.	Use case ends

3.4.7 Input data

Table 19 -Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Receiver Name	Name of recipient	Yes	String	Nguyen Van A
2.	Phone Number	Recipient phone	Yes	10 digits and starts with "0"	0912345678

3.	Province	Location for fee calculation	Yes	Selected from list	Ha Noi
4.	Address	Delivery address	Yes	String	1 Dai Co Viet
5.	Shipping instructions	Additional delivery notes	No	String	Deliver in office hours
6.	Payment Method	Choice of QR code or Credit Card (Default QR code)	Yes	QR code or Paypal	VietQR

3.4.8 Output data

Table 20 Output data of invoice view

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product	String	DVD Phim A
2.	Price	Unit price of corresponding product	Right alignment, positive integer and comma for thousands separator	60,000
3.	Quantity	Quantity ordered of corresponding product	Right alignment, positive integer and comma for thousands separator	2
4.	Total product price excluding VAT	Total for the corresponding product without VAT	Right alignment, positive integer and comma for thousands separator	120,000
5.	Total product price including VAT	Total amount of all products with VAT	Right alignment, positive integer and comma for thousands separator	120,000
6.	Shipping Fee	Calculated shipping cost	Right alignment, positive integer and comma for thousands separator	50,000
7.	Total	Final amount	Right alignment, positive integer and comma for thousands separator	170,000

Table 21 -Output data of general order and transaction information

No	Data fields	Description	Display format	Example
1.	Customer name	Name of recipient	String	Nguyen Van A

2.	Phone number	Recipient phone	10 digits and starts with "0"	0912345678
3.	Province	Location for fee calculation	Selected from list	Ha Noi
4.	Shipping address	Delivery address	String	246,000
5.	Total amount	Final paid amount	Right alignment, positive integer and comma for thousands separator	170,000
6.	Transaction ID	Unique identifier	String	123456
7.	Transaction content	Description of transaction	String	Order 456 payment
8.	Transaction datetime	Date and time of the transaction	dd/mm/yyyy hh:mm:ss	05/10/2023 11:31:23

3.4.9 Postconditions

- **Success:** The order is recorded as "Pending," payment is confirmed, and stock is updated.
- **Failure:** No order is recorded; the system remains in its previous state.

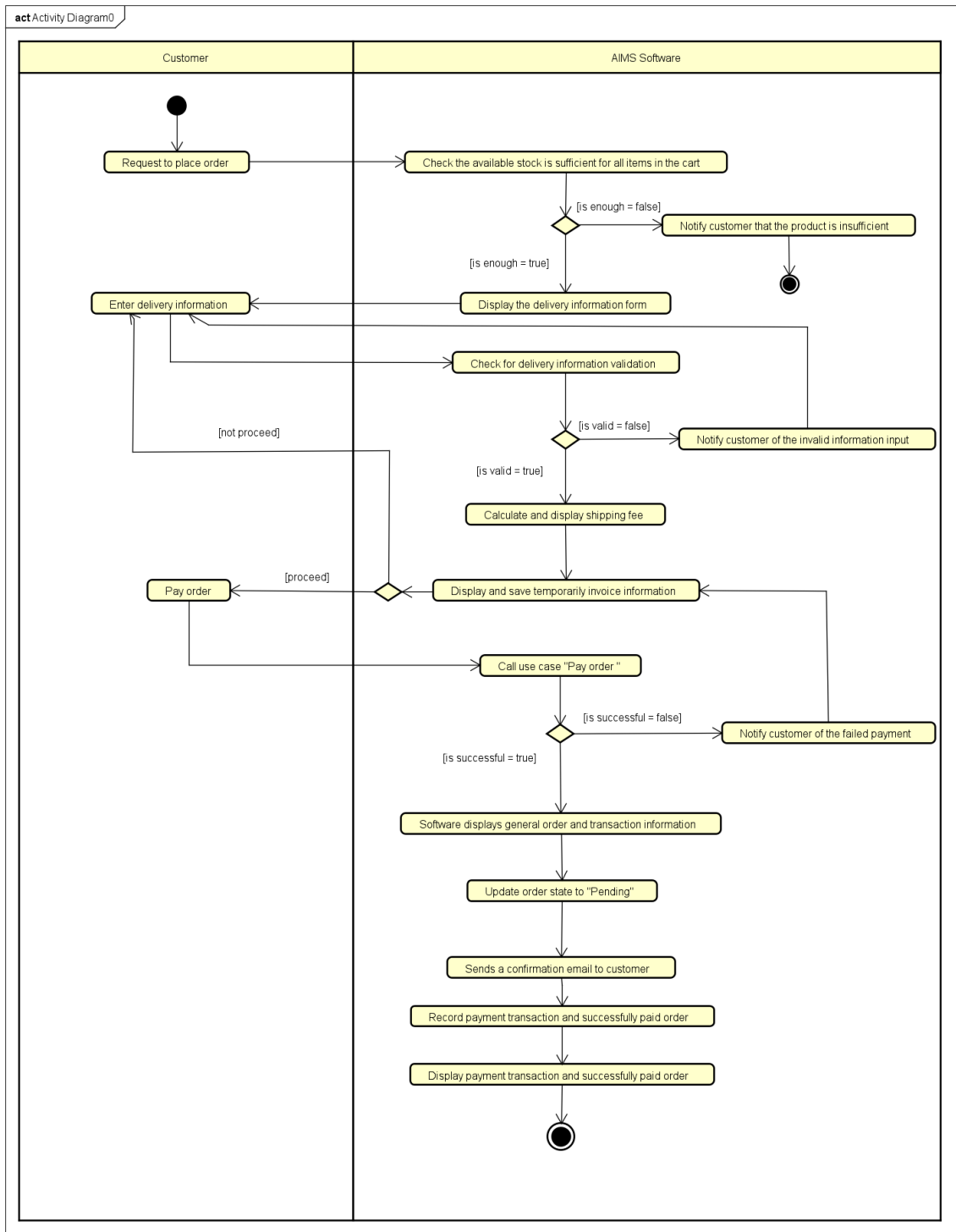


Figure 6: Activity Diagram for Use Case Place Order

3.5 Use Case “Pay Order”

3.5.1 Use case code

UC005

3.5.2 Brief Description

This usecase describes the interaction between a customer and the AIMS software when completing payment for an order using the default VietQR payment method. The customer scans a QR code displayed by the system to complete the payment through their banking application. The system processes the payment through the VietQR external payment gateway, updates the invoice status upon successful payment, records the transaction information.

3.5.3 Actors

- Customer
- VietQR

3.5.4 Preconditions

- Order information is available and valid
- Delivery information has been provided and valid
- Stock availability has been verified for all products in the cart

3.5.5 Basic Flow of Events

1. AIMS Software displays the payment screen with invoice information (list of products, quantities, prices, total product price excluding VAT, total product price including VAT, delivery fee, total amount to be paid)
2. AIMS Software calls the VietQR API to generate a QR code for the payment
3. VietQR returns the generated QR code to AIMS Software
4. AIMS Software displays the QR code on the payment screen together with the invoice information
5. Customer scans the QR code using their banking application and completes the transfer
6. VietQR sends a payment callback to AIMS Software indicating successful payment
7. AIMS Software updates the invoice status to reflect successful payment
8. AIMS Software saves a new order with the associated invoice in the system

3.5.6 Alternative flows

Table 22: Alternative flows of events for UC Pay Order

No	Location	Condition	Action	Resume location
1.	At Step 1	Customer chooses to pay with Credit Card	▪ AIMS Software switch to Credit Card payment process	End
2.	At Step 6	VietQR sends a payment callback to AIMS Software	▪ AIMS Software displays payment failure message to the customer.	Resumes at Step 1

		indicating payment failure	- Customer chooses to attempt payment again or switch to alternative payment method. - AIMS Software proceeds to appropriate payment flow based on customer choice.	
3.	At Step 2	AIMS Software attempts to call VietQR API but the service is unavailable or returns an error	- AIMS Software displays error message indicating payment service is temporarily unavailable. - Customer chooses to retry later or switch to alternative payment method. - AIMS Software proceeds based on customer choice.	Resumes at Step 1
4.	At Step 4	Customer does not complete the payment within the allowed time period	- AIMS Software detects that payment has not been completed within the timeout period and displays timeout message to the customer. - Customer chooses to retry payment or switch to alternative payment method. - AIMS Software proceeds based on customer choice.	Resumes at Step 1
5.	Any Step	Customer cancels the paying process	AIMS Software return to the main page of the software	End

3.5.7 Input data

Table 23: Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Customer Name	Name of the customer placing the order	Yes	Non-empty string, valid characters	"Nguyen Van A"
2.	Phone Number	Contact phone number for delivery	Yes	Valid Vietnamese phone number format	"0912345678"
3.	Shipping Address	Delivery address for the order	Yes	Non-empty string, valid address format	"123 Nguyen Trai, Hanoi"
4.	Product List	List of products in the cart	Yes	At least one product, all products in stock	Book, CD, DVD, Newspaper
5.	Product Quantity	Quantity of each product in the cart	Yes	Positive integer, not exceeding available stock	2

6.	Product Price	Price of each product (excluding VAT)	Yes	Between 30% and 150% of original value	150000
7.	Total Product Price (excl. VAT)	Sum of all product prices excluding VAT	Yes	Calculated value, positive number	300000
8.	Total Product Price (incl. VAT)	Total product price with 10% VAT	Yes	Calculated value, positive number	330000
9.	Delivery Fee	Shipping cost based on weight and location	Yes	Calculated according to shipping rules, max 25000 VND for orders over 100000 VND	22000
10.	Total Amount	Final amount to be paid (incl. VAT + delivery fee)	Yes	Calculated value, positive number	352000

3.5.8 Output data

Table 24: Output data

No	Data fields	Description	Display format	Example
1.	Invoice Information	Complete invoice details including products, quantities, prices, VAT, delivery fee, and total amount	Table format with itemized list	Table showing all order details
2.	Payment Status	Current status of the payment transaction	Text message (Success/Failure/Timeout)	"Success"
3.	Order Information	General order details after successful payment	Text format with customer name, phone, address, province, total amount	"Customer: Nguyen Van A, Phone: 0912345678, Address: 123 Nguyen Trai, Hanoi, Total: 352000 VND"
4.	Transaction ID	Unique identifier for the payment transaction	Alphanumeric string	"TXN123456789"

5.	Transaction Content	Description of the transaction content	Text string describing order details	"Payment for Order #12345"
6.	Transaction Datetime	Timestamp of successful payment	Date and time format (YYYY-MM-DD HH:MM:SS)	"2026-03-15 15:30:00"
7.	Error Message	Description of payment failure or error	Clear, actionable text message	"Payment failed. Please try again or use alternative payment method."

3.5.9 Postconditions

- Invoice status is updated to indicate successful payment
- A new order is saved in the system with the associated invoice
- Order is in "pending processing" state awaiting product manager approval
- Invoice and payment transaction information is sent to the customer's email

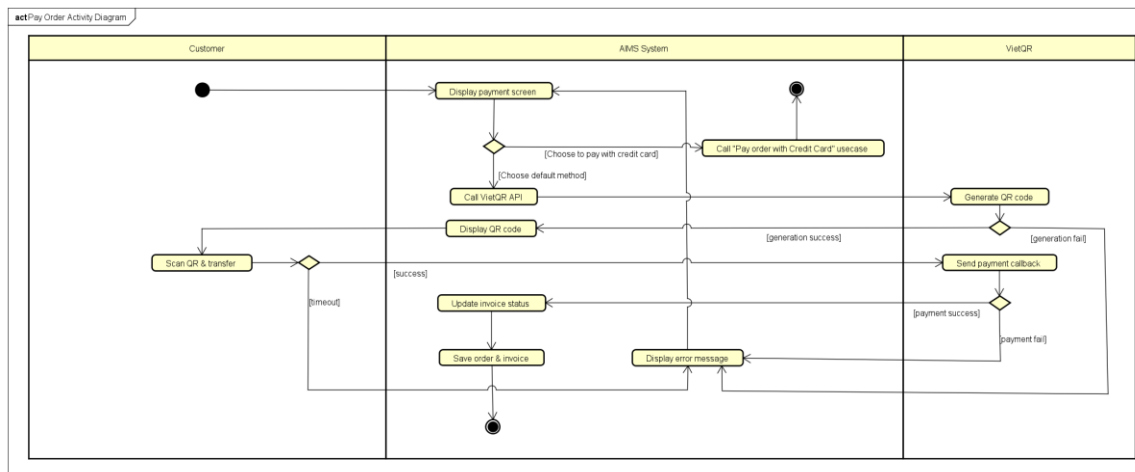


Figure 7: Activity Diagram for Pay Order

3.6 Use Case “Pay order by Credit Card”

3.6.1 Use case code

UC006

3.6.2 Brief Description

This use case describes the interaction between a customer and the AIMS software when completing payment for an order using the **Credit Card payment method through PayPal**. The system redirects the customer to the PayPal payment gateway where the customer enters credit card information to complete the payment. PayPal processes the payment transaction and returns the result to the AIMS system. The system updates the invoice status upon successful payment and records the transaction information.

3.6.3 Actors

- **Customer:** The customer who initiates the payment using a credit card.
- **PayPal Gateway:** External payment gateway that processes the credit card payment and returns the payment result to AIMS.

3.6.4 Preconditions

- Customer selected Credit Card payment

3.6.5 Basic Flow of Events

1. AIMS Software creates a payment request and sends it to the PayPal API with order information and total amount.
2. PayPal returns a response for payment processing.
3. AIMS Software redirects the Customer to the PayPal payment interface.
4. PayPal displays the credit card payment screen.
5. Customer enters credit card information
 - cardholder name
 - card number
 - expiration date
 - CVV
6. Customer confirms the payment.
7. PayPal validates and processes the payment transaction.
8. PayPal returns the payment result to AIMS Software.
9. AIMS Software updates the invoice status to reflect successful payment
10. AIMS Software saves a new order with the associated invoice in the system

3.6.6 Alternative flows

Table 25: N-Alternative flows of events for UC Pay by Credit Card

No	Location	Condition	Action	Resume location
1.	At Step 1	AIMS Software attempts to call PayPal API but the service is unavailable or returns an error	▪ AIMS Software displays error message indicating payment service is temporarily unavailable.	Use Case ends

2.	At Step 6	Customer does not complete the payment within the allowed time (timeout)	- PayPal cancels the payment session. - AIMS Software displays timeout message. - Customer may retry or choose another payment method.	Use Case ends
	At Step 8	Payment fails after processing	- AIMS Software notifies Customer that payment failed. - Customer may retry payment or switch method.	Use Case ends

3.6.7 Input data

Table 26: Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Order ID	Unique identifier of the order to be paid	Yes	Must exist in system	"ORD123456"
2.	Total Amount	Total amount to be paid (including VAT and shipping fee)	Yes	Positive number, matches invoice	170000
3.	Payment Method	Selected payment method	Yes	Must be "Credit Card"	Credit Card

3.6.8 Output data

Table 27: Output data

No	Data fields	Description	Display format	Example
1.	Invoice Information	Complete invoice details including products, quantities, prices, VAT, delivery fee, and total amount	Table format with itemized list	Table showing all order details

2.	Payment Status	Current status of the payment transaction	Text message (Success/Failure/Cancelled)	"Success"
3.	Order Information	General order details after successful payment	Text format with customer name, phone, address, province, total amount	"Customer: Nguyen Van A, Phone: 0912345678, Address: 123 Nguyen Trai, Hanoi, Total: 352000 VND"
4.	Transaction ID	Unique identifier for the payment transaction	Alphanumeric string	"TXN987654321"
5.	Transaction Content	Description of the transaction content	Text string describing order details	"Payment for Order #12345"
6.	Transaction Datetime	Timestamp of successful payment	Date and time format (YYYY-MM-DD HH:MM:SS)	"2026-03-15 15:45:00"
7.	Error Message	Description of payment failure or error	Clear actionable message	"Credit card payment failed. Please try again or use another payment method."

3.6.9 Postconditions

- Invoice status is updated to indicate successful payment
- A new order is saved in the system with the associated invoice
- Order is in **"pending processing"** state awaiting product manager approval
- Invoice and payment transaction information is sent to the customer's email

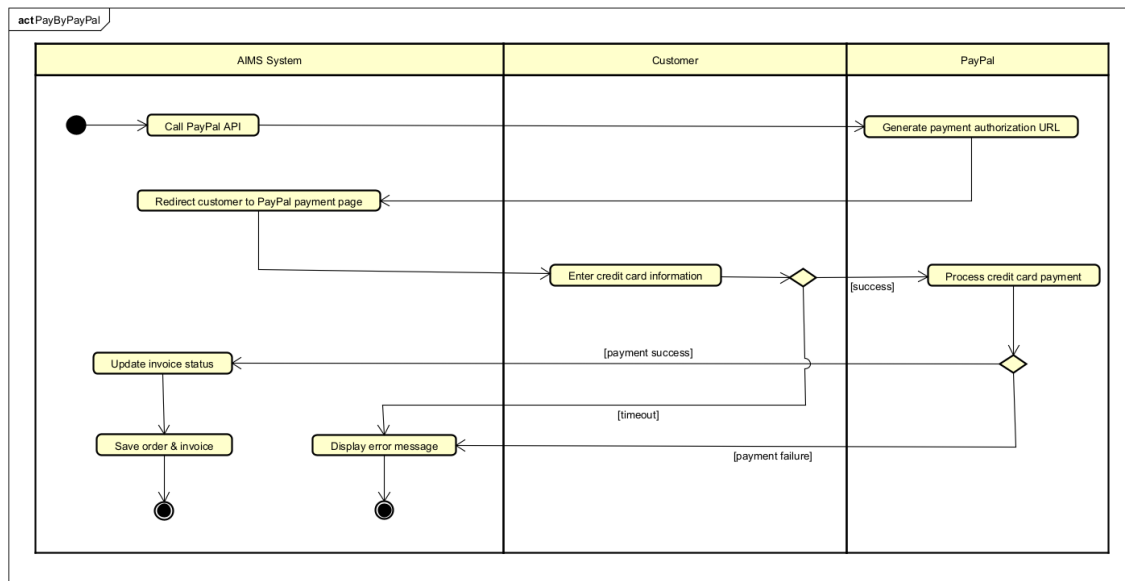


Figure 8: Activity Diagram – Pay Order by Credit Card

3.7 Use Case “View Product”

3.7.1 Use case code

UC007

3.7.2.Brief Description

Allows users to view detailed information about a selected product.

3.7.3. Actors

- Customer
- Product Manager

3.7.4. Preconditions

- The user is on the product list screen.
- Products exist in the system.

3.7.5 Basic Flow of Events

1. User starts software
2. System shows a list of 20 random products
3. User selects a product from the list
4. AIMS Software retrieves the product information from the database
5. Aims software displays general information of the product
6. AIMS Software displays the detailed information of the selected product depending on the product type
7. User views the product details. The usecase ends

3.7.6 Alternative flows

Table 28-Alternative flows of events for UC View Product Detail

No	Location	Condition	Action	Resume location
•	At Step 5	If the actor is Product Manager	System displays additional management information.	Use case ends
•	At Step 4	If the system cannot retrieve product information due to a system error	System displays an error message and asks the user to try again later.	Use case ends

3.7.7.Input data

Table 29: Input data

No	Data fields	Description	Mandatory	Valid condition	Example
----	-------------	-------------	-----------	-----------------	---------

•	Product ID	ID of the product	Yes	Must exist in the database	67
•	Product Type	Type of the product used to determine how to display the information	Yes	Must be Customer or Product Manager	Book, DVD, CD,...
•	User Role	Role of the actor accessing the product detail	Yes	Non-empty string, valid address format	Customer

3.7.8. Output data

Table 30-Output data of Book Product Information

No	Data fieldss	Description	Display format	Example
1	Title	Name of the book	Text string	"Lion King"
2	Category	Product category	Text string	"Books"
3	Value	Original value of the book	Currency format (VND)	"100,000 VND"
4	Price	Selling price of the book	Currency format (VND)	"150,000 VND"
5	Authors	Author(s) of the book	Text string	"To Hoai"
6	Cover type	Type of book cover (paperback or hardcover)	Text string	"Hardcover"
7	Publisher	Publishing company	Text string	"NXB Nhi Dong"
8	Publication date	Date the book was published	dd/mm/yyyy	"20/02/2025"
9	Additional information	Additional details such as number of pages, language, or genre	Text string	"520 pages"

Table 31-Output Data of CD Product Information

No	Data fields	Descriptions	Display format	Example
1	Title	Name of the CD or album	Text string	"Chung Ta"
2	Category	Product category	Text string	"CDs"
3	Value	Original value of the CD	Currency format (VND)	"200,000 VND"
4	Price	Selling price of the CD	Currency format (VND)	"250,000 VND"
5	Artists	Artist(s) performing the album	Text string	"Son Tung MTP"
6	Record label	Company producing the CD	Text string	"MTP Entertainment"
7	Tracklist	List of tracks in the CD, each including title and duration	Text list	"Con Mua Ngang Qua – 4:30, Lac Troi – 3:55"
8	Genre	Music genre of the CD	Text string	"Pop"
9	Additional information	Additional details such as release date	Text string	"Released in 2017"

Table 32-Output Data – DVD Product Information

No	Data fields	Description	Display format	Example
1	Title	Name of the DVD	Text string	"Lion King"
2	Category	Product category	Text string	"DVDs"
3	Value	Original value of the DVD	Currency format (VND)	"300,000 VND"
4	Price	Selling price of the DVD	Currency format (VND)	"780,000 VND"
5	Disc type	Type of disc (Blu-ray or HD-DVD)	Text string	"Blu-ray"
6	Director	Director of the film	Text string	"Jon Favreau"
7	Runtime	Total duration of the film	hh:mm:ss	"01:20:04"
8	Studio	Film production studio	Text string	"Marvel Studios"
9	Language	Audio language of the DVD	Text string	"English"

Table 33-Output Data – Newspaper Product Information

No	Data fields	Description	Display format	Example
1	Title	Name of the newspaper	Text string	"Tuoi Tre Daily"
2	Category	Product category	Text string	"Newspapers"
3	Value	Original value of the newspaper	Currency format (VND)	"10,000 VND"
4	Price	Selling price of the newspaper	Currency format (VND)	"12,000 VND"
5	Editor-in-chief	Person responsible for editorial content	Text string	"Nguyen Van A"
6	Publisher	Organization publishing the newspaper	Text string	"Tuoi Tre Publishing House"
7	Publication date	Date the newspaper issue was published	dd/mm/yyyy	"20/02/2025"
8	Issue number	Identification number of the newspaper issue	Numeric / Text	"Issue 245"
9	Publication frequency	Frequency of publication (daily, weekly, etc.)	Text string	"Daily"

3.7.9.Postconditions

- Product details are displayed.

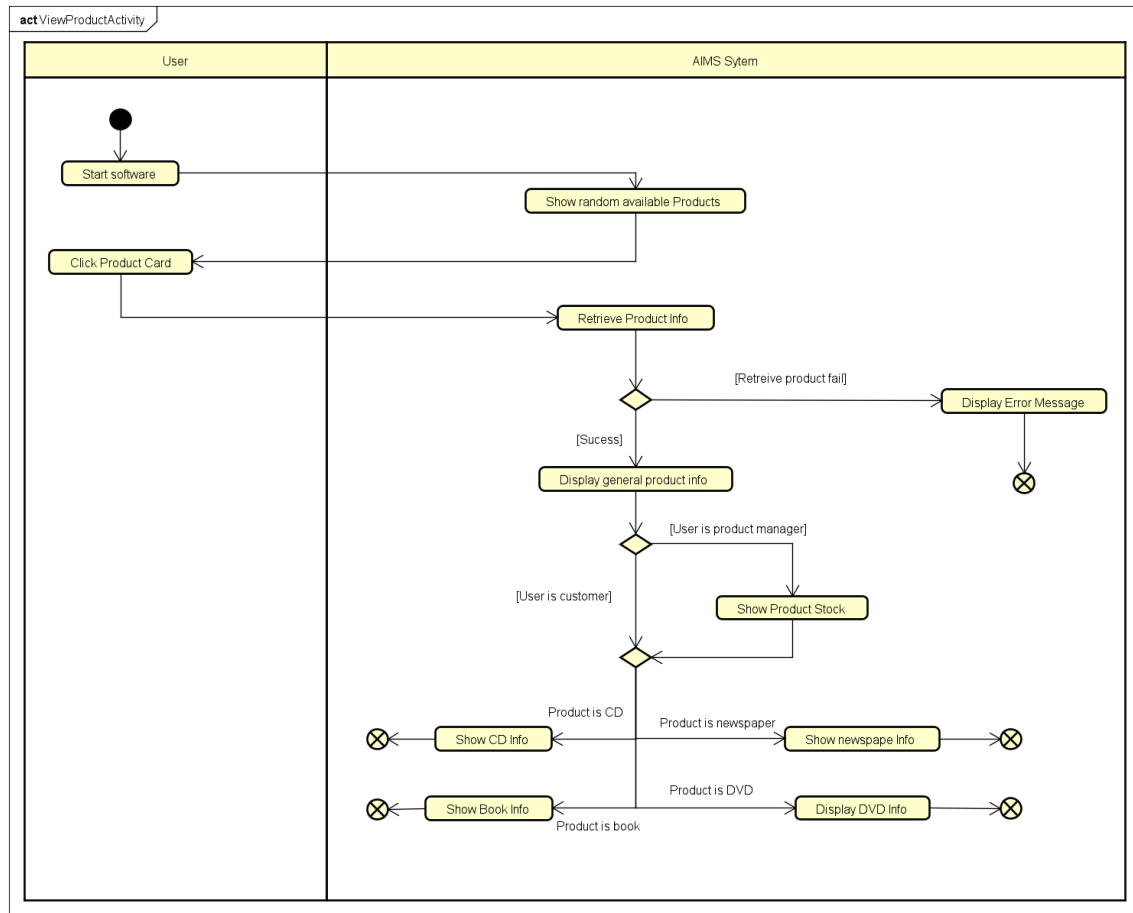


Figure 9: Activity Diagram – View Product

4 Supplementary specification

4.1 Functionality

- **Audit Logging:** The system shall maintain a permanent history of all product-related actions (additions, edits, deletions), recording the Manager ID, timestamp, and a description of the specific changes made.
- **Transaction Recording:** The software must record all payment transaction details (ID, Content, Datetime) and store successfully paid orders in a secure audit log for historical tracking.
- **Real-time Validation & Notifications:**
 - The system shall notify users of invalid operations, including input non-compliance, incorrect date formats, or exceeding batch/daily limits.
 - **Stock Alerts:** Users must be notified immediately if the requested quantity in the cart exceeds available stock.
- **Batch Operation Constraints:** The system shall enforce a business rule where Add/Edit operations are limited to one product at a time, while Delete operations allow for a maximum of 10 products simultaneously.
- **Pricing Integrity:** For all product modifications, the system shall enforce a business rule where the current price must remain between **30% and 150%** of the product's original value.
- **Inventory Management:**
 - Stock quantity shall be automatically updated upon completed sales.
 - Manual adjustments by Product Managers require a recorded reason for auditing purposes.
- **Automated Refunds:**
 - **PayPal:** The system shall process refunds automatically upon order cancellation (before approval) or rejection by a manager.
 - **VietQR:** The system shall notify the Product Manager to perform a manual refund outside the system.
- **Administrative & Security Alerts:** All sensitive administrative actions (account creation, blocking, role modifications) shall be logged, and affected users must be notified automatically via email.
- **Automated Communications:** The system shall automatically send invoices and transaction summaries to customers upon successful payment, as well as status updates (Approval/Rejection) following manager reviews.

4.2 Usability

- **User Familiarization:** The interface shall be designed intuitively so that new users can perform basic tasks (searching, cart management) within minutes without prior training.

- **Public Accessibility:** Customers shall be able to view detailed information for all product types (Books, CDs, DVDs, Newspapers), search the catalog, and apply price filters without requiring authentication.
- **Discovery & Search:**
 - Upon startup, the system shall display a list of 20 random products to encourage exploration.
 - The search interface shall support querying by title and category with responsive filtering by price range.
- **Cart Informative Feedback:** The system shall provide immediate, clear feedback to the user, such as notifying them specifically if stock is insufficient for a requested quantity or if a form field contains invalid data (e.g., incorrect email format).
- **Checkout Context Preservation:** During the delivery setup, the system shall allow customers to view their shopping cart contents simultaneously to ensure context is not lost during the checkout process.
- **Order Review Pagination:** To ensure ease of use for managers, the pending order review interface shall display a maximum of 30 orders per page, allowing for efficient navigation through large volumes of pending requests.
- **Administrative Access Management:** Administrators shall have a clear and efficient interface to trigger secure password resets for staff accounts without having visibility of the underlying password data.

4.3 Reliability

- **System Availability:** The platform shall operate as an e-commerce service available 24/7 to customers.
- **Failure-Free Operation:** The system shall maintain continuous service for a minimum of 300 hours without a critical system failure.
- **Incident Recovery:** The software must be capable of resuming normal operations within a maximum of 1 hour following a service interruption or incident.
- **Data Consistency (Stock Integrity):** The system shall guarantee that stock quantity are updated atomically upon a successful transaction, ensuring that the same physical media item cannot be sold twice simultaneously.
- **Email Delivery Reliability:** The system shall ensure that transaction summaries and invoices are successfully dispatched to the customer's email immediately after the "Pending Processing" state is reached.

4.4 Performance

- **Concurrent User Support:** The system shall support up to 1,000 simultaneous customers without any detectable degradation in performance.

- **Response Times:**
 - Under normal conditions, maximum response time shall be 2 seconds.
 - During peak hours, response time shall not exceed 5 seconds.

4.5 Supportability

- **Platform Flexibility:** The system shall be developed to run as both a responsive web application and a cross-platform desktop application.
- **Auditable Maintenance:** Transaction logs must include Manager IDs and specific change deltas to facilitate efficient system auditing and troubleshooting.
- **Development & Testing Safety:** Integration with payment gateways (PayPal, VietQR) shall utilize Sandbox environments to ensure safe testing and demonstration without real financial risk.
- **Product Types Extensibility:** The system architecture shall be designed to allow for the future addition of new media types without requiring a rewrite of the core inventory logic.
- **Gateway Extensibility:** The architecture shall support modular integration of future payment gateways beyond the initial PayPal and VietQR implementations.

4.6 Other requirements

- **Taxation Standards:**
 - All stored and displayed prices (Original and Current) shall be **exclusive** of a 10% Value-Added Tax (VAT).
 - The system shall calculate and display the VAT separately on temporary invoices and final receipts.
- **Shipping Fee Rules (Tax-Exempt):** Shipping fees are not subject to VAT and are calculated based on weight and destination:
 - **Hanoi & HCM City:** 22,000 VND for the first 3kg.
 - **Other Provinces:** 30,000 VND for the first 0.5kg.
 - **Incremental Weight:** An additional 2,500 VND for every subsequent 0.5kg.
 - **Free Shipping Promotion:** Orders with a total item value exceeding 100,000 VND qualify for a shipping discount (Free Shipping) capped at a maximum of 25,000 VND per order.
- **Security & Data Protection:**
 - **Password Integrity:** All staff passwords must be securely hashed using industry-standard algorithms.
 - **Access Control:** Administrators shall not have access to retrieve or view raw password data for any user.
 - **Self-Service Security Management:** The system shall provide a secure interface for all staff (Admins and Managers) to update their own passwords independently at any time.

- **Administrative Transparency:** Administrators shall not have access to retrieve or view raw password data for any user at any time.
- **Access Control Policies:**
 - User management shall follow the principle of least privilege.
 - The system must support **Role Multiplicity**, allowing a single user to be assigned multiple roles (e.g., Administrator and Product Manager simultaneously).
- **Strict Deletion Constraints:**
 - Permanent deletion is only permitted for products with zero stock; products with remaining stock must be "deactivated" instead.
 - **Security Limit:** Each manager is restricted to a maximum of 20 product deletions per day.
- **Refund Policy Details:**
 - **Automated:** PayPal refunds are processed automatically by the system upon cancellation or rejection.
 - **Manual:** VietQR refunds must be handled manually by the Product Manager outside the system, though the system will provide the necessary transaction details for manual processing.